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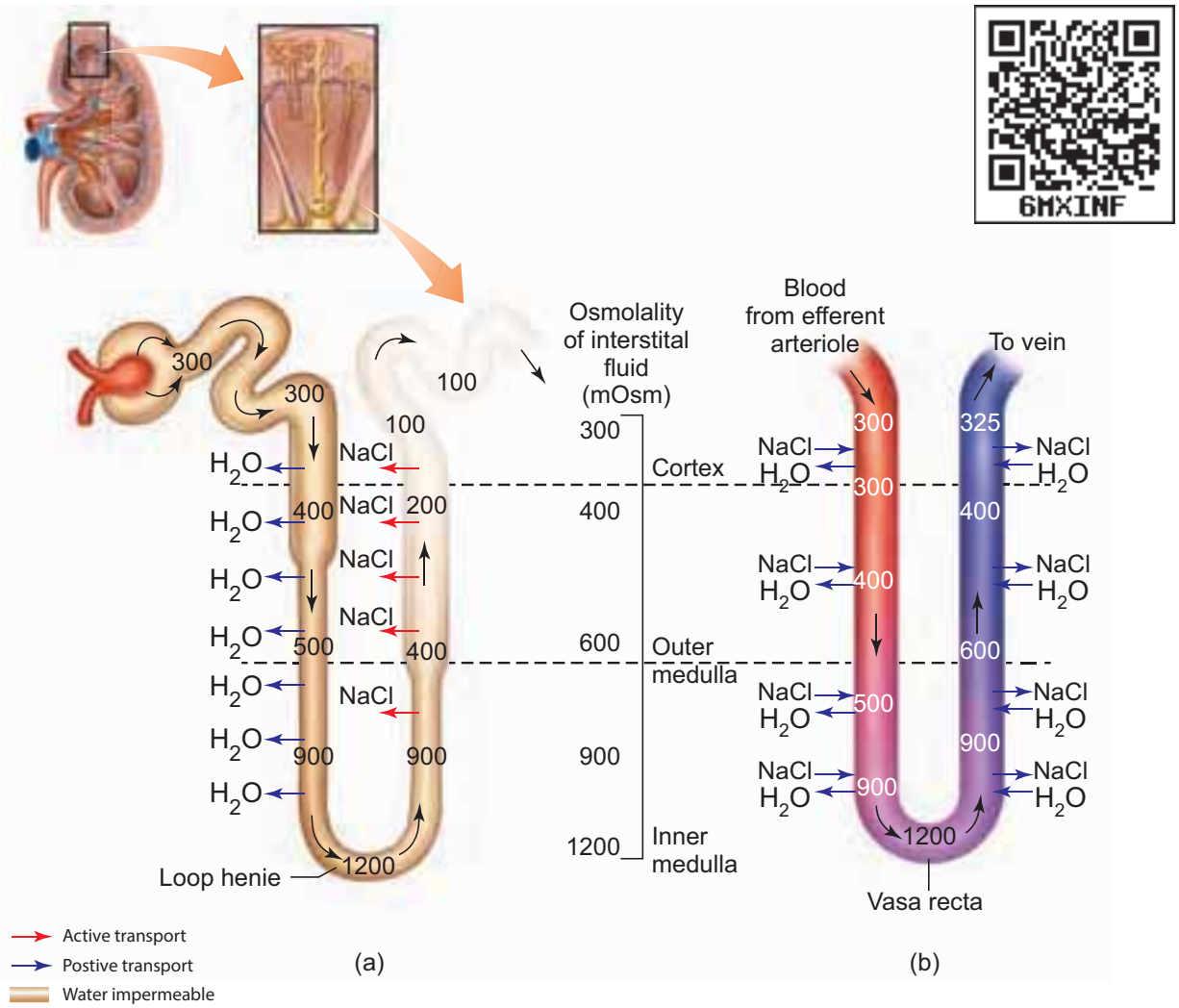


Figure 8.10 (a) Counter current multiplier – the long loops of Henle of the juxtamedullary nephrons create the medullary osmotic gradient (b) Counter current exchanger – Vasa recta preserves the medullary gradient while removing reabsorbed water and solutes.

current mechanisms. The major function of Henle's loop is to concentrate Na^1 and Cl^2. There is low osmolarity near the cortex and high osmolarity towards the medulla. This osmolarity in the medulla is due to the presence of the solute transporters and is maintained by the arrangement of the loop of Henle, collecting duct and vasa recta. This arrangement allows movement of solutes from the filtrate to the interstitial fluid. At the transition between the proximal convoluted tubule and the descending loop of Henle the osmolarity of the interstitial fluid is similar to that of the blood – about 300mOsm. Ascending and descending limbs of Henle, create a **counter current multiplier** (interaction between flow of filtrate through the limbs of Henle's and JMN) by active transport. Figure 8.10 (a) shows the counter current multiplier created by the long loops of Henle of the JM nephrons which creates medullary osmotic gradient. As the fluid enters the descending limb, water moves from the lumen into the interstitial fluid and the osmolarity decreases. To counteract this dilution the region of the